

Macroeconomic Determinants of U.S. Housing Prices

Evidence from U.S. Monthly Data (1990–2025)

Author: Saeed Janghorban (sd.janghorban@gmail.com)

1. Project Overview

This project investigates the relationship between macroeconomic factors and U.S. housing price growth using monthly data from 1990 to 2025.

The primary objective is to examine whether key macroeconomic variables, including interest rates, unemployment, and inflation, can explain changes in housing price dynamics.

Various economic conditions influence housing markets. Interest rates affect borrowing costs and mortgage affordability; unemployment reflects household income uncertainty and labor market conditions; and inflation represents broader price movements in the economy.

This project applies Ordinary Least Squares (OLS) regression and conducts several diagnostic tests to evaluate the reliability of the estimated model.

2. Data Description

The dataset consists of monthly U.S. macroeconomic indicators collected from the Federal Reserve Economic Database (FRED). The variables used in the analysis are:

Source	Description	Variable
FRED	S&P/Case-Shiller U.S. National Home Price Index	Housing Price Index
FRED	Federal Funds Rate	Interest Rate
FRED	U.S. Unemployment Rate	Unemployment Rate
FRED	CPI-based annual inflation rate	Inflation

The sample period covers: January 1990 – December 2025

After merging the datasets and removing missing observations, the final dataset was used for regression analysis.

3. Data Preparation and Variable Construction

The original housing price index is transformed into a monthly growth rate to analyze changes rather than absolute price levels. The dependent variable is defined as:

$$HousingGrowth_t = \frac{Housing\ Price_t - Housing\ Price_{t-1}}{Housing\ Price_{t-1}} \times 100$$

Inflation is calculated using a year-over-year percentage change in the Consumer Price Index:

$$Inflation_t = \frac{CPI_t - CPI_{t-12}}{CPI_{t-12}} \times 100$$

4. Exploratory Data Analysis

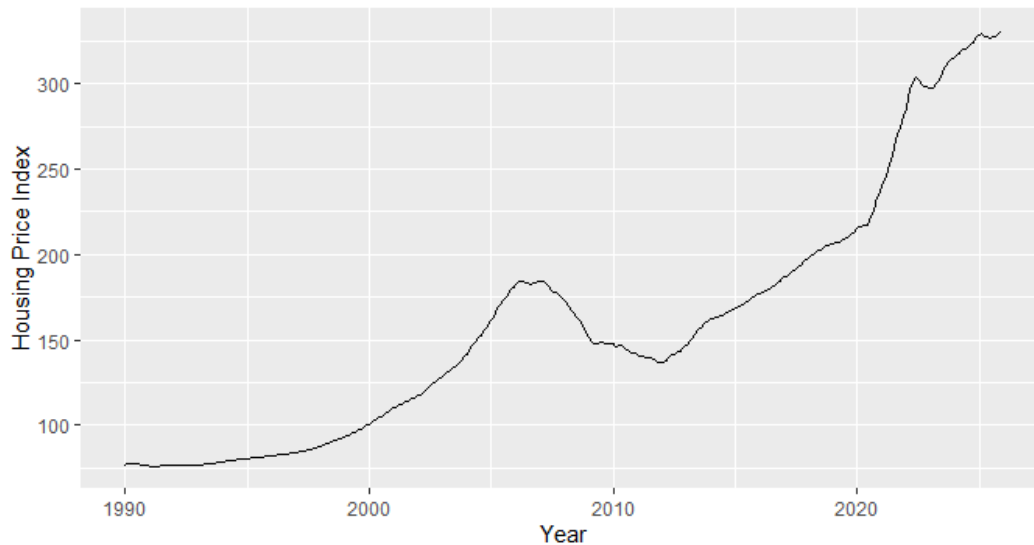
4.1 Housing Price Trend

The housing price index shows a strong upward trend between 1990 and 2025.

The most notable movements include:

- Rapid appreciation before the 2007–2008 financial crisis.
- A significant decline during the housing market crash.
- A strong recovery after 2012.
- A sharp increase during and after the COVID-19 period.

U.S. Housing Price Index (1990-2025)



4.2 Descriptive Statistics

Descriptive statistics were calculated to understand the distribution and characteristics of the variables. The analysis includes:

- Mean
- Standard deviation
- Minimum and maximum values
- Distribution characteristics

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
housing_price	2	431	158.7756148	71.52966355	148.087	149.5509623	71.5265544	75.564	331.288	255.724	0.869309403	0.015842	3.445462656
interest_rate	3	431	2.876450116	2.345187189	2.61	2.756173913	3.602718	0.05	8.29	8.24	0.251541394	-1.22171	0.112963692
unemployment	4	431	5.677262181	1.741716054	5.4	5.453913043	1.63086	3.4	14.8	11.4	1.269057149	2.071952	0.08389551
cpi	5	431	209.9946775	51.52457413	211.398	206.6245565	58.4055444	127.5	326.031	198.531	0.390727879	-0.68887	2.481851406
housing_growth	6	430	0.34172308	0.546520825	0.342557378	0.354712999	0.387047884	-1.615857564	1.904062421	3.519919985	-0.27844867	1.08286	0.026355575
inflation	7	419	2.625047733	1.547364208	2.592165899	2.506961774	1.040498658	-1.958761004	8.979361103	10.93812211	1.115355466	3.488244	0.075593688

4.3 Correlation Analysis

A correlation matrix was calculated to examine the relationship between explanatory variables and the housing market.

	housing_price	interest_rate	unemployment	cpi
housing_price	1	-0.255909217	-0.331449833	0.957338886
interest_rate	-0.255909217	1	-0.445239113	-0.42413244
unemployment	-0.331449833	-0.445239113	1	-0.209149287
cpi	0.957338886	-0.42413244	-0.209149287	1

Correlation analysis provides preliminary insights but does not imply causal relationships.

5. Econometric Model Specification

The baseline econometric model is estimated using Ordinary Least Squares (OLS). The model examines how macroeconomic variables influence housing price growth.

$$HousingGrowth_t = \beta_0 + \beta_1 InterestRate_t + \beta_2 Unemployment_t + \beta_3 Inflation_t + \epsilon_t$$

Where:

- ***HousingGrowth_t*** = represents monthly housing price growth.
- ***InterestRate_t*** = represents the federal funds rate.
- ***Unemployment_t*** = represents the unemployment rate.
- ***Inflation_t*** = represents CPI-based inflation.
- ***ε_t*** = represents the error term.

6. Regression Results

The OLS regression estimates the relationship between housing price growth and macroeconomic variables.

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.259374151	0.13256931	9.499741289	1.70E-19
interest_rate	-0.078873654	0.013082448	-6.02896737	3.65E-09
unemployment	-0.120627492	0.016697385	-7.224334167	2.43E-12
inflation	-0.002416577	0.017416475	-0.138752333	0.889713166

The estimated coefficients are interpreted based on their signs, magnitude, and statistical significance.

The main findings are:

- Interest rates show the expected negative relationship with housing price growth, suggesting that higher borrowing costs may reduce housing demand.
- Unemployment demonstrates a negative relationship with housing growth, indicating that weaker labor market conditions reduce housing market activity.
- Inflation does not show a statistically significant relationship with housing price growth in the estimated model.

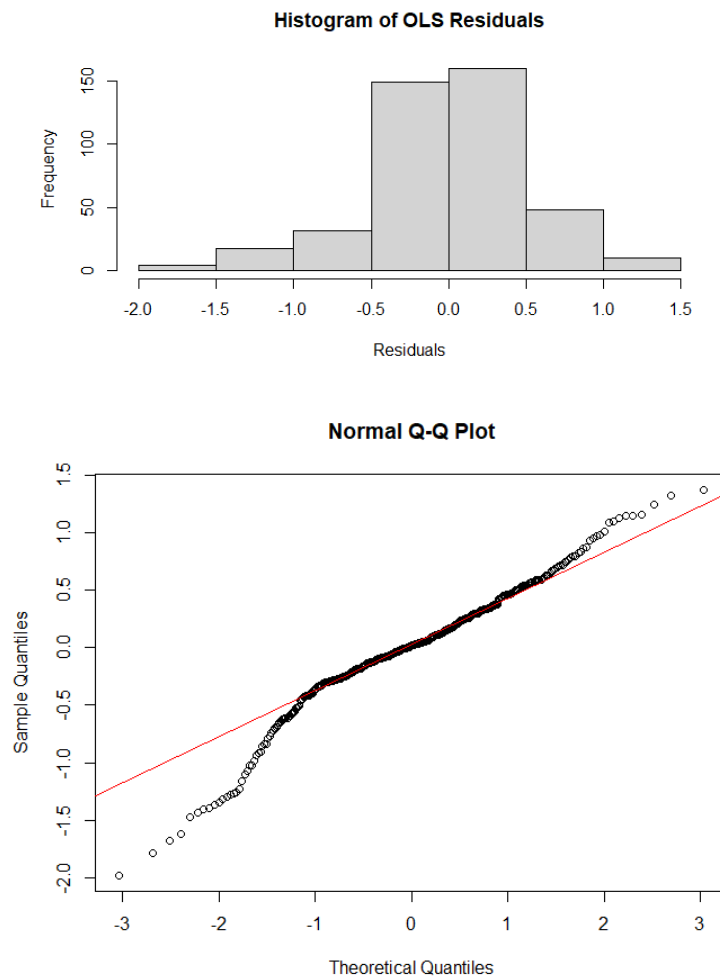
7. Model Diagnostic Tests

To evaluate the validity of the OLS assumptions, several diagnostic tests were performed.

7.1 Residual Normality

Residual normality was evaluated using:

- Histogram analysis
- Q-Q plot
- Jarque-Bera test



The Jarque-Bera test indicates that residuals deviate from a normal distribution. This suggests that the normality assumption of classical OLS errors is violated.

7.2 Heteroskedasticity Test

The Breusch-Pagan test was conducted to examine whether the variance of residuals is constant.

```
BP = 52.291, df = 3, p-value = 2.596e-11
```

The test detects heteroskedasticity, indicating that the classical OLS standard errors may affect the reliability of conventional statistical inference.

7.3 Autocorrelation Test

Since the dataset is a time series, residual autocorrelation was tested using:

- Durbin-Watson test
- Breusch-Godfrey test

```
lag Autocorrelation D-W Statistic p-value
1      0.9242725      0.1506706      0
Alternative hypothesis: rho != 0
```

The results indicate significant positive autocorrelation in residuals.

7.4 Multicollinearity Test

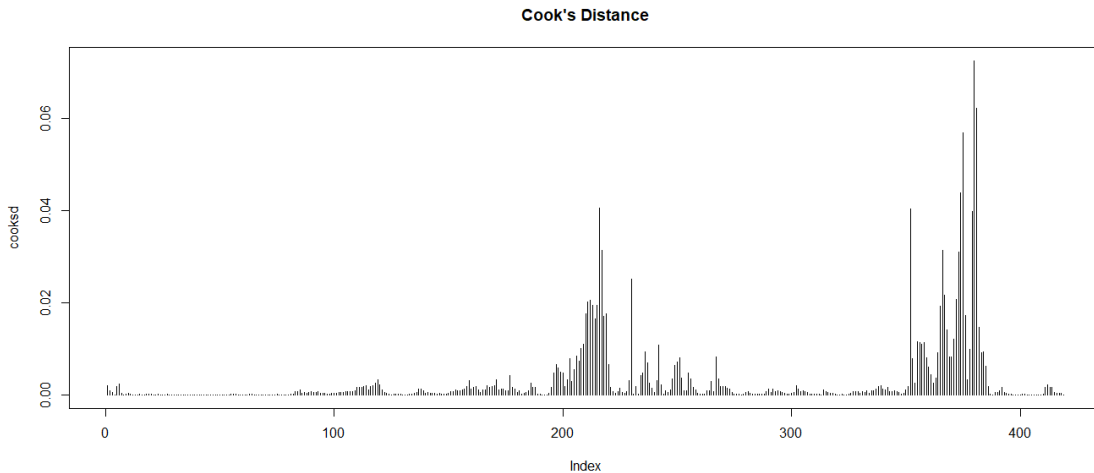
Variance Inflation Factor (VIF) was calculated to detect possible multicollinearity among explanatory variables.

```
interest_rate unemployment inflation
1.321336      1.383167      1.156050
```

All VIF values are below conventional thresholds, indicating that multicollinearity is not a concern in the model.

7.5 Influential Observations

Cook's Distance was used to identify observations with excessive influence on regression estimates.



Several observations show stronger influence. These points were reviewed but not removed because they may represent economically meaningful periods. However, no observation was removed because these events represent economically meaningful periods.

8. Model Performance

The explanatory power of the model was evaluated using:

- R-squared
- Adjusted R-squared
- F-test

R_squared	Adjusted_R_squared
0.132939451	0.126671543

The R-squared value indicates the proportion of variation in housing price growth explained by the included macroeconomic variables.

The F-test evaluates whether the explanatory variables are jointly significant.

9. Conclusion

This project analyzes the macroeconomic determinants of U.S. housing price growth using monthly data from 1990 to 2025.

The main conclusions are:

1. Labor market conditions, particularly unemployment, play an important role in explaining housing price movements.
2. Interest rates have the theoretically expected negative relationship with housing growth, although their statistical significance is limited in the baseline specification.
3. Inflation does not provide strong additional explanatory power in the baseline specification.

4. Diagnostic tests reveal violations of some classical OLS assumptions, including heteroskedasticity and autocorrelation. These findings indicate potential limitations of the baseline model specification.

Overall, the analysis demonstrates how macroeconomic conditions influence housing markets and provides an empirical framework for studying housing price dynamics.